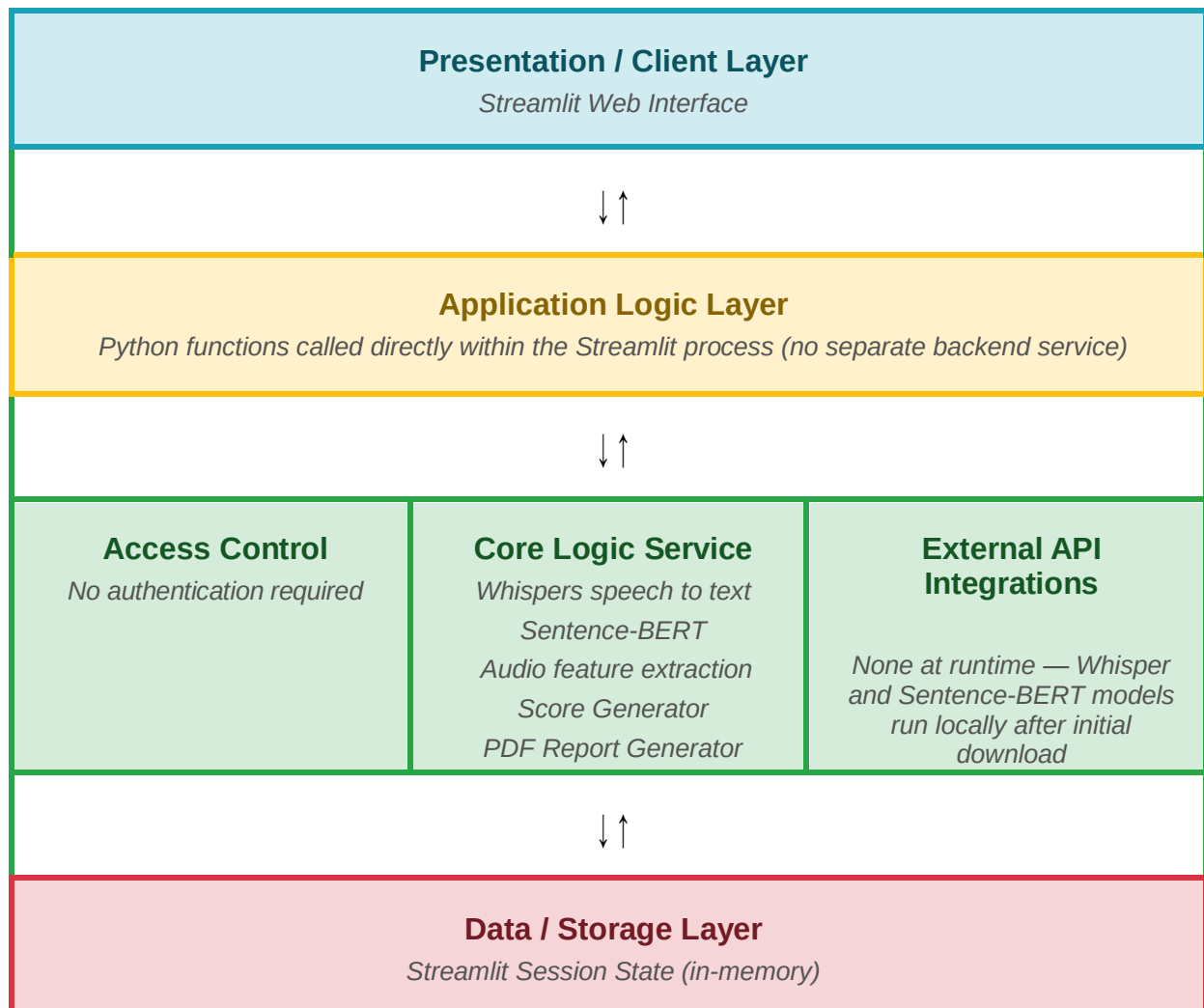


Solution Architecture

Date	1 July 2026
Team ID	SWTID-2026-9706
Project Name	Voice Based Concept understanding Analyser
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>User interface for recording/uploading audio and union results</i>	<i>Streamlit</i>
<i>Application Logic</i>	<i>Calls core logic functions directly within the same Streamlit process</i>	<i>Python</i>
<i>Core Logic</i>	<i>Speech recognition, Semantic Analysis, scoring, report generation</i>	<i>Whisper, Sentence-BERT, Librosa, ReportLab</i>
<i>Data Layer</i>	<i>Stores report, audio, and reference concepts</i>	<i>Streamlit Session State</i>